

[1]: !pip install matplotlib

```
Requirement already satisfied: matplotlib in c:\users\sai\appdata\local\programs\python\python313\lib\site-pack
Requirement already satisfied: contourpy>=1.0.1 in c:\users\sai\appdata\local\programs\python\python313\lib\si
lib) (1.3.3)
Requirement already satisfied: cyclers>=0.10 in c:\users\sai\appdata\local\programs\python\python313\lib\site-pa
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\sai\appdata\local\programs\python\python313\lib\si
tlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\sai\appdata\local\programs\python\python313\lib\si
tlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\sai\appdata\local\programs\python\python313\lib\site-pac
(2.4.1)
Requirement already satisfied: packaging>=20.0 in c:\users\sai\appdata\local\programs\python\python313\lib\site
ib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\sai\appdata\local\programs\python\python313\lib\site-packag
2.1.0)
Requirement already satisfied: pyparsing>=3 in c:\users\sai\appdata\local\programs\python\python313\lib\site-pac
(3.3.1)
```

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Requirement already satisfied: python-dateutil<2.7 in c:\users\sai\appdata\local\programs\python\python313\lib\site-packages (2.9.0.post0)

Requirement already satisfied: six>=1.5 in c:\users\sai\appdata\local\programs\python\python313\lib\site-packages (1.17.0)

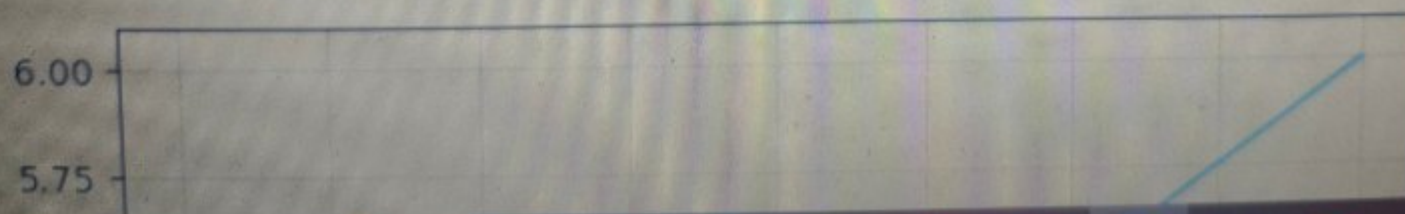
```
[2]: import matplotlib.pyplot as plt
import pandas as pd
```

```
[3]: # 1. Basic Example
```

```
x = [1,2,3]
```

```
y = [4,5,6]
```

```
[5]: plt.plot(x,y)
plt.grid()
plt.show()
```



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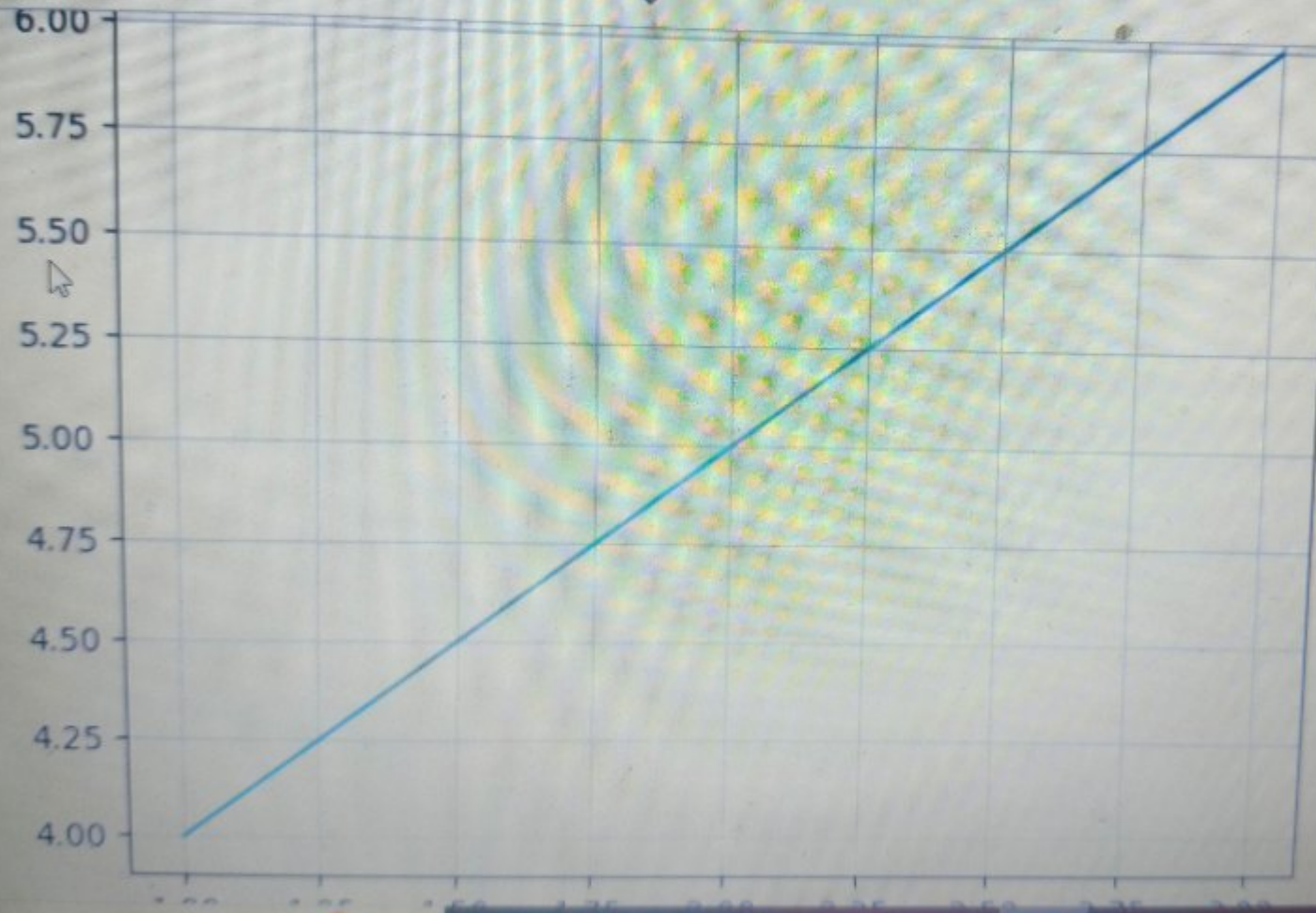
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[6]: # 2 . Pyplot API

[7]: #univariate

[8]: data = {
 "Salary" : [25000 , 30000, 24000, 50000, 23000, 45000 , 100000, 100000, 30000]
}

[9]: df = pd.DataFrame(data)

[10]: df.head()

[11]:

	Salary
0	25000
1	30000
2	24000
3	50000



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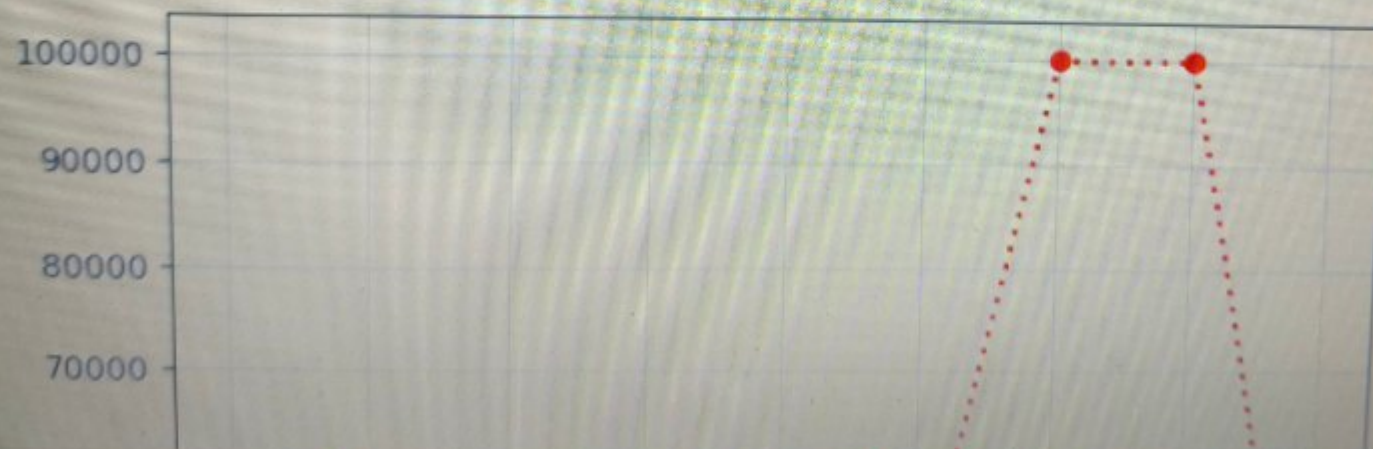
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[12]: df.shape

[12]: (9, 1)

[14]: # 3 . LinePlot

```
plt.plot(df["Salary"], color = "red", marker = "o", linestyle = ":", linewidth = "2")  
plt.grid()  
plt.show()
```

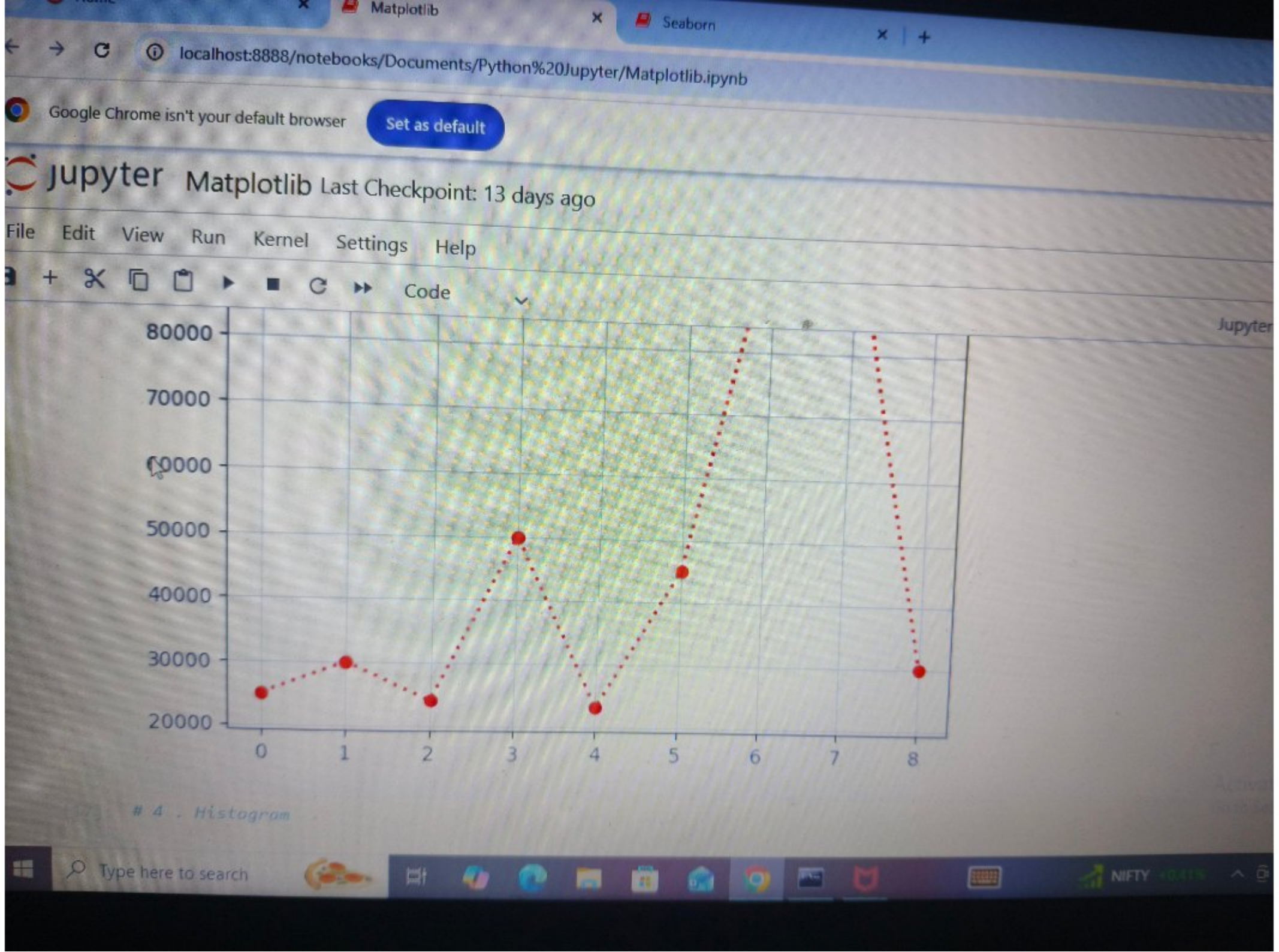


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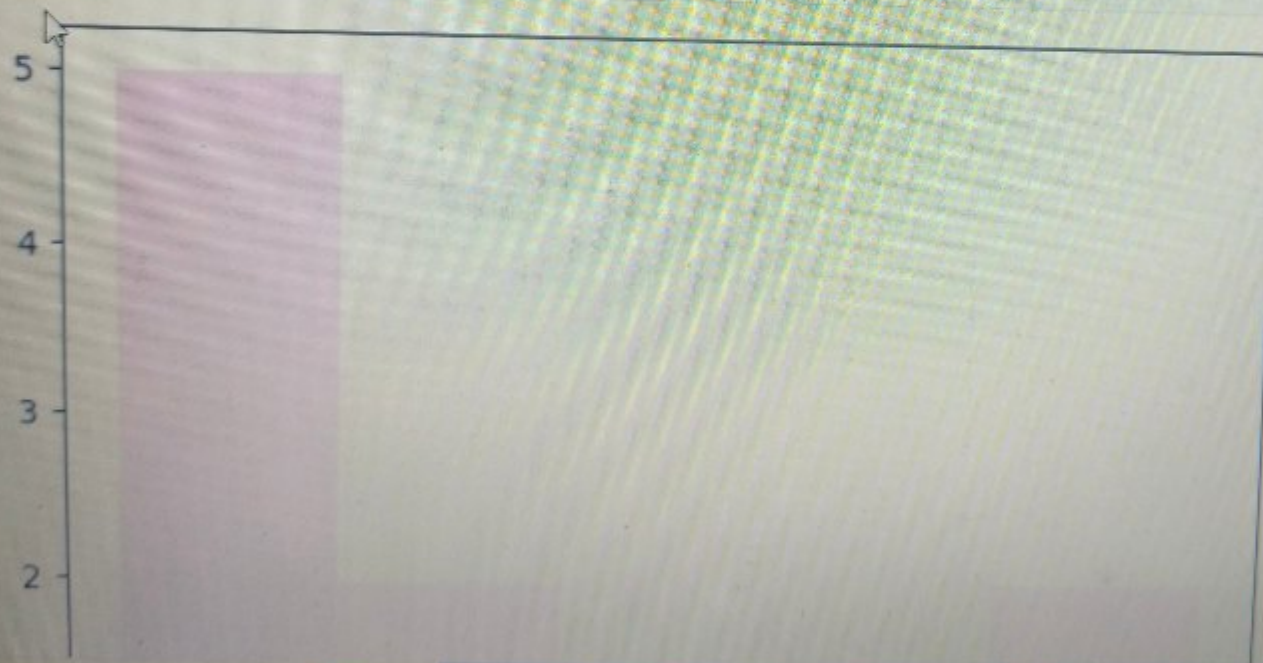
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[17]: # 4 . Histogram

```
plt.hist(df["Salary"], bins = 5, color = "pink")  
plt.show()
```



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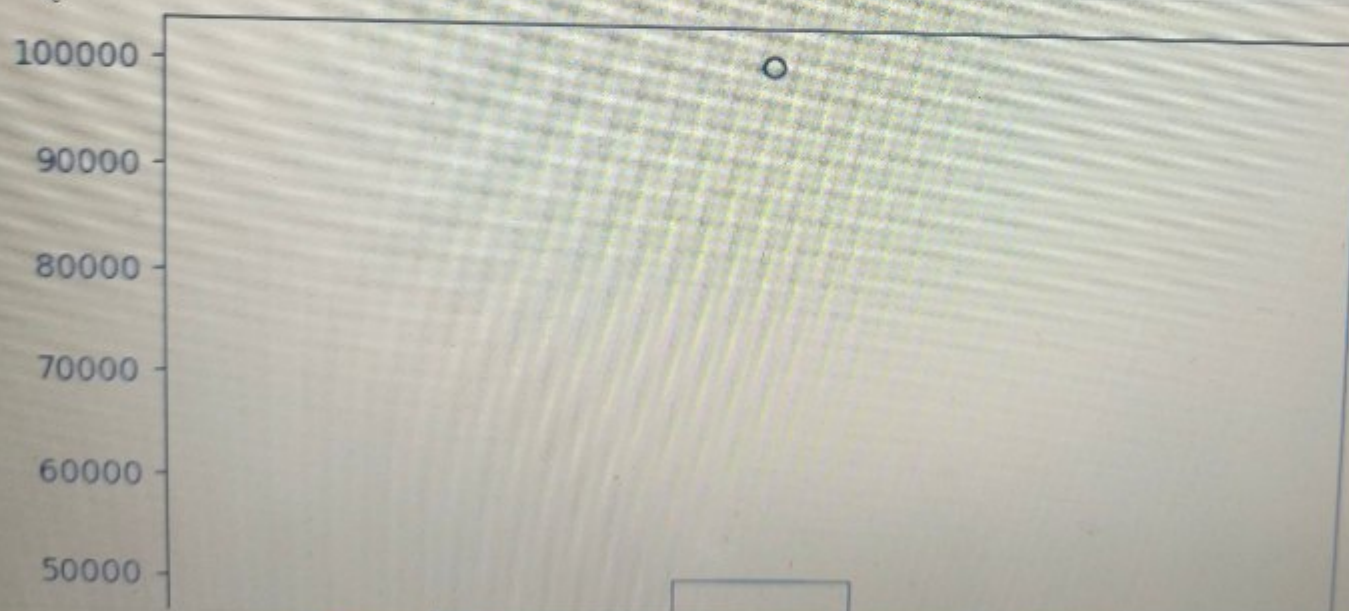
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[18]: # 5 . BoxPlot

```
plt.boxplot(df["Salary"])  
plt.show()
```



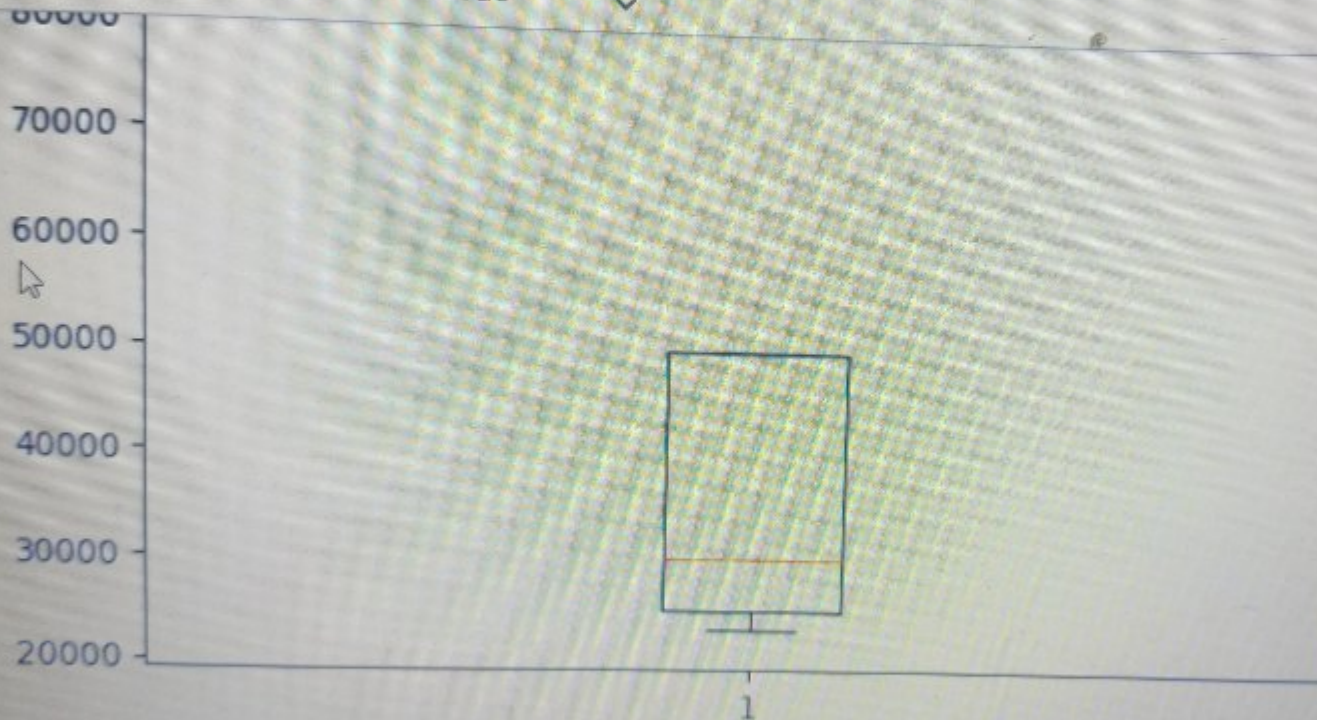
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Ollama version 0.4.5.6

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Code



```
[19]: # Univariate : categorical
```

```
df["dept"] = ['Hr', 'dev', 'Finanace', 'HR', 'fullStack', 'WebDev', 'HR', 'AIDev', 'Merndev']
```

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Code

```
[19]: # Univariate : categorical
```

```
[23]: df["dept"] = ['Hr' , 'dev', 'Finanace' , 'HR', 'fullStack' , 'WebDev', 'HR', 'AIDev' , 'Merndev' ]
```

```
[24]: df.head()
```

```
[24]:
```

	Salary	dept
0	25000	Hr
1	30000	dev
2	24000	Finanace
3	50000	HR
4	23000	fullStack

```
[25]: df.info()
```



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```
[25]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  ---
0   Salary  9 non-null        int64
1   dept    9 non-null        object
dtypes: int64(1), object(1)
memory usage: 276.0+ bytes
```

[26] # 6. PieCharts

```
count = df["dept"].value_counts()
count
```

[26] dept
HR
Hr
dev
si

localhost:8888/notebooks/Documents/Python%20Jupyter/Matplotlib.ipynb



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memory usage: 276.0+ bytes

Jupyter

[26]: # 6. PieCharts

```
count = df["dept"].value_counts()
count
```

[26]: dept

HR	2
Hr	1
dev	1
Finanace	1
fullStack	1
WebDev	1
AIDev	1
Merndev	1

Name: count, dtype: int64

```
[27]: plt.pie(count, labels = count.index, autopct = "%1.2f")
plt.axis("equal")
plt.show()
```



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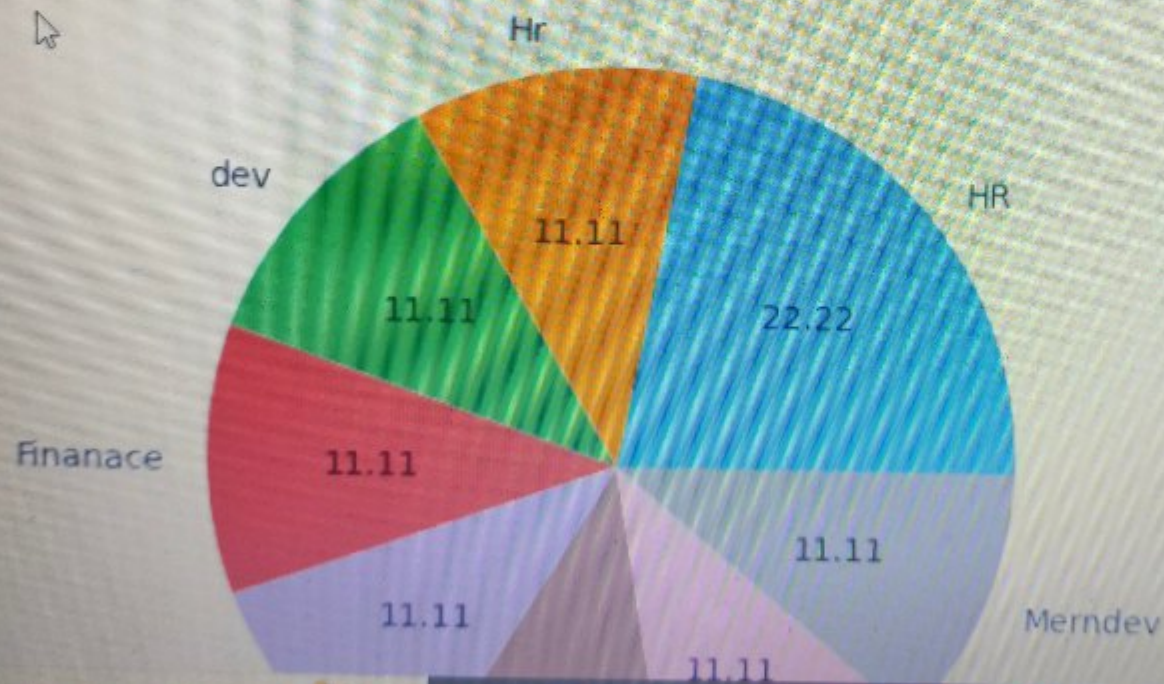
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Code

```
[37]: plt.pie(count, labels = count.index, autopct = "%1.2f")  
plt.axis("equal")  
plt.show()
```



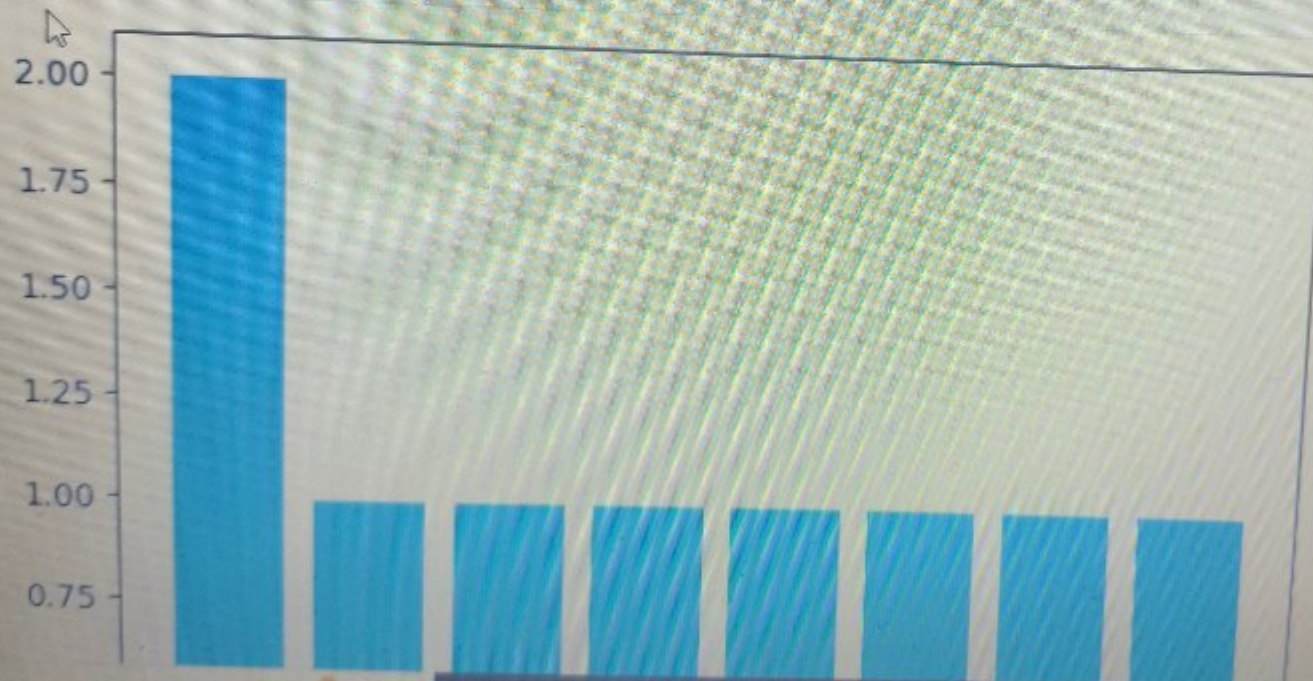
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[38]: #7 . countPlot

```
plt.bar(count.index , count)
plt.show()
```



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```
[40]: #8 Bivariate Analysis : numerical-numeical
```

```
[41]: df["Age"] = [20, 22, 23, 24, 23, 22, 21, 23, 22]
      df.head()
```

```
[41]:
```

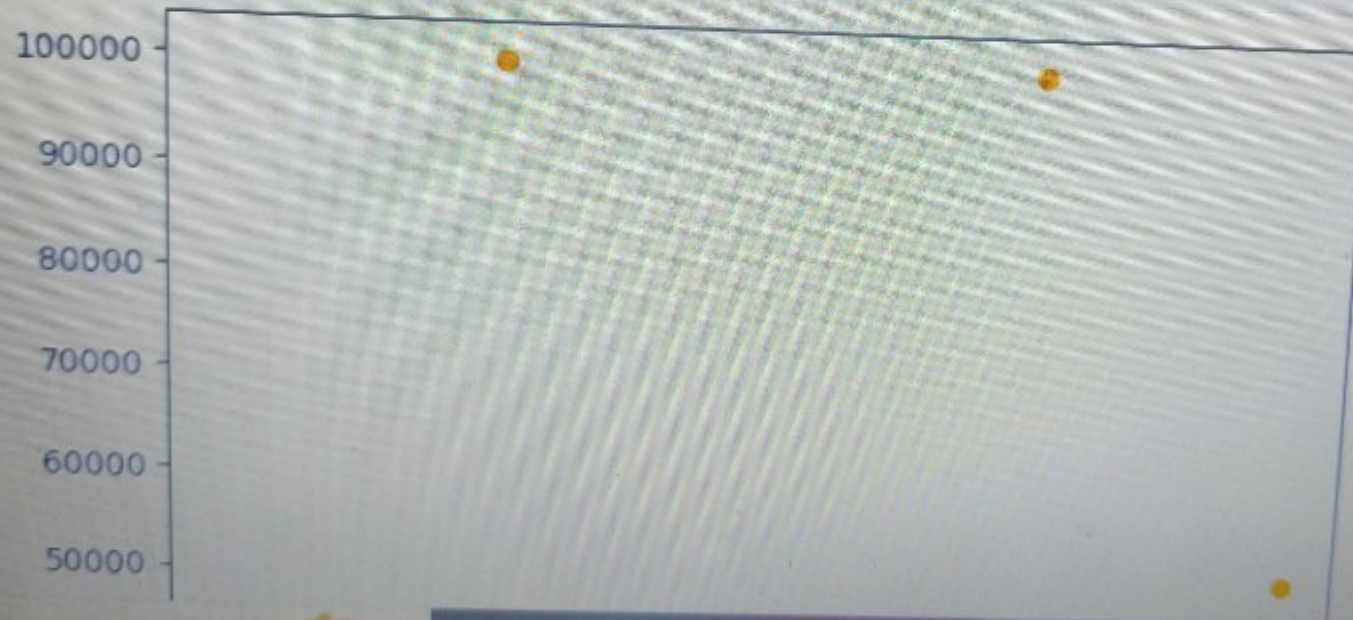
	Salary	dept	Age
0	25000	Hr	20
1	30000	dev	22
2	24000	Finanace	23
3	50000	HR	24
4	23000	fullStack	23

```
[42]: #9 . Scatter plot
```

```
plt.scatter(df["Age"], df["Salary"], color = "orange")
```

[42]: #9 . Scatter plot

```
plt.scatter(df["Age"], df["Salary"], color = "orange")  
plt.show()
```



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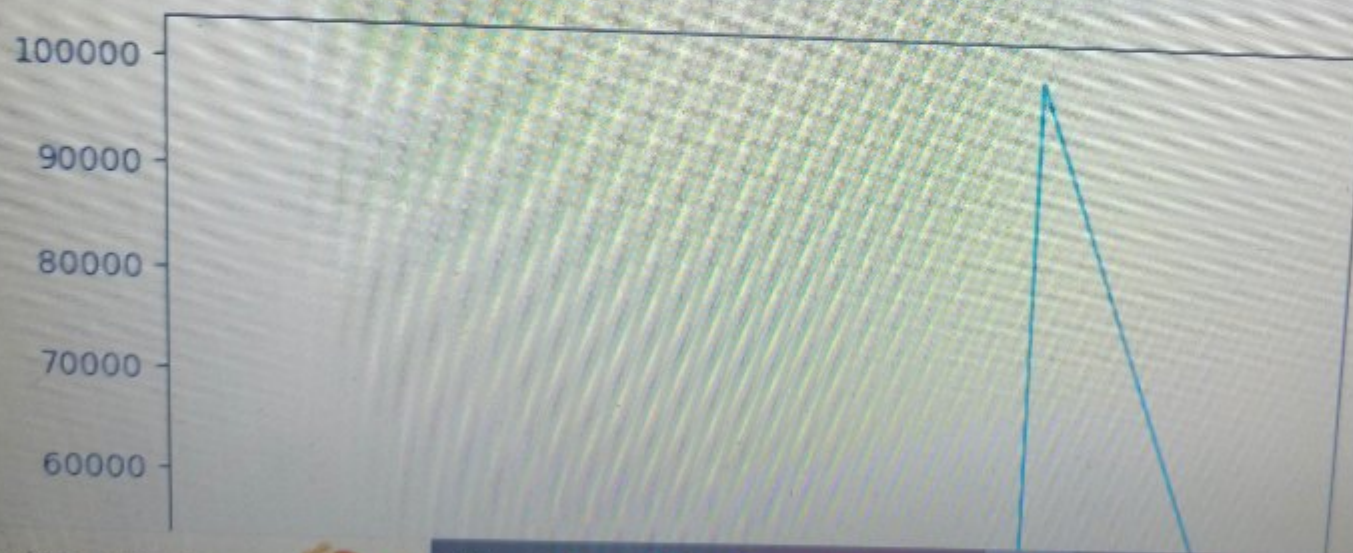
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```
[45]: sort_age = df.sort_values("Age")
```

```
[46]: # lineplot
```

```
plt.plot(sort_age["Age"], df["Salary"])  
plt.show()
```



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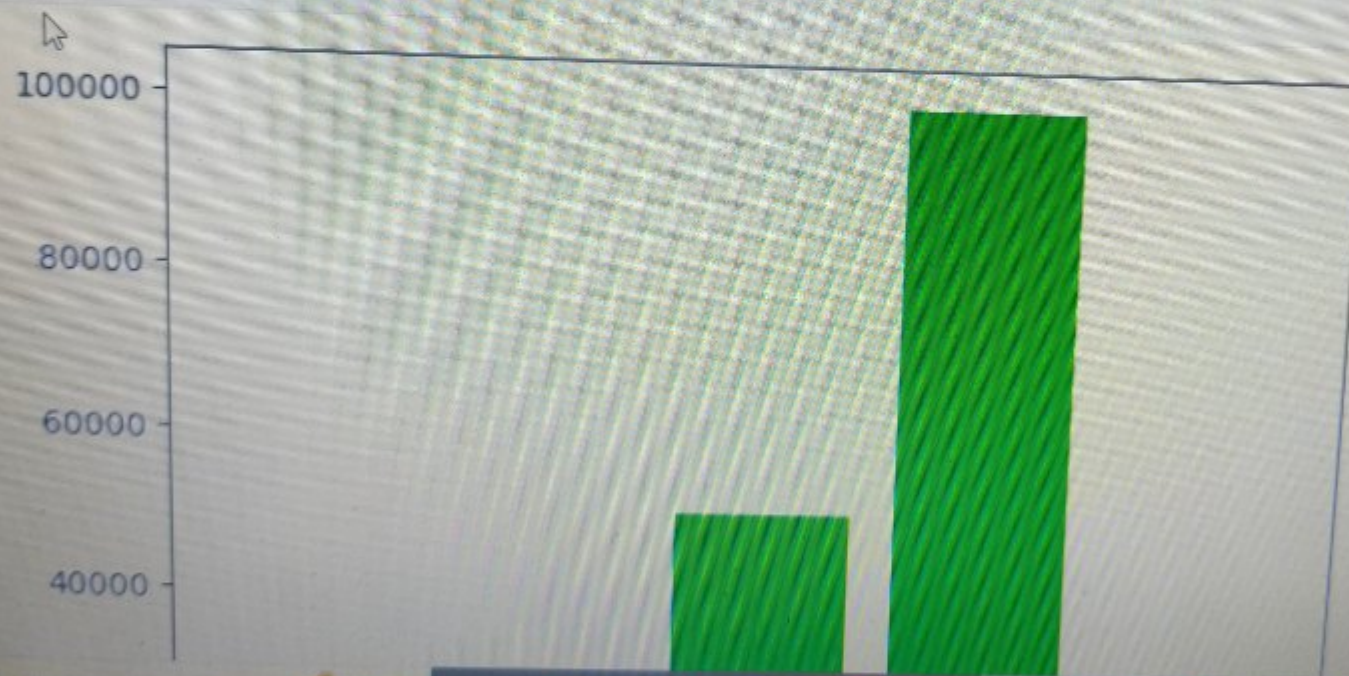


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[47]: #10. Barchart

```
plt.bar(sort_age["Age"], df["Salary"], color = "green")  
plt.show()
```





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Code

20

21

22

23

24

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```
[48]: # 11 Bivariate : Numerical ; categorical
```

```
[50]: hr_sal = df[df["dept"] == "HR"]["Salary"]  
hr_sal
```

```
[50]: 3    50000  
6    100000  
Name: Salary, dtype: int64
```

```
[52]: dev_sal = df[df["dept"] == "dev"]["Salary"]  
dev_sal
```

```
[52]: 1    30000  
Name: Salary, dtype: int64
```

```
[53]: #Boxplot
```

```
plt.boxplot([hr_sal , dev_sal])  
plt.show()
```



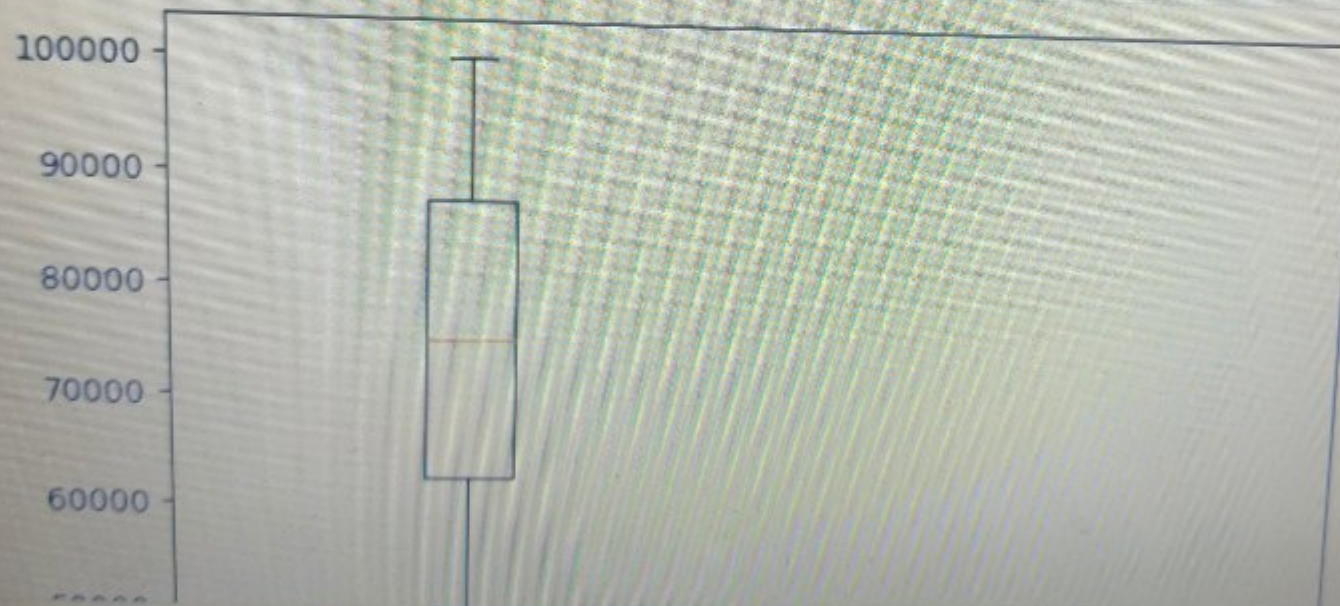
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[54]: #Boxplot

```
plt.boxplot([hr_sal, dev_sal])  
plt.show()
```





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Code

[57]: #piechart

```
salary_by_dept = df.groupby("dept")["Salary"].sum()
salary_by_dept
```

[57]:

```
dept
AIDev      100000
Finanace    24000
HR          150000
Hr          25000
Merndev     30000
WebDev      45000
dev         30000
fullStack   23000
Name: Salary, dtype: int64
```

```
[58]: plt.pie(salary_by_dept, autopct = "%1.2f")
plt.axis("equal")
plt.show()
```



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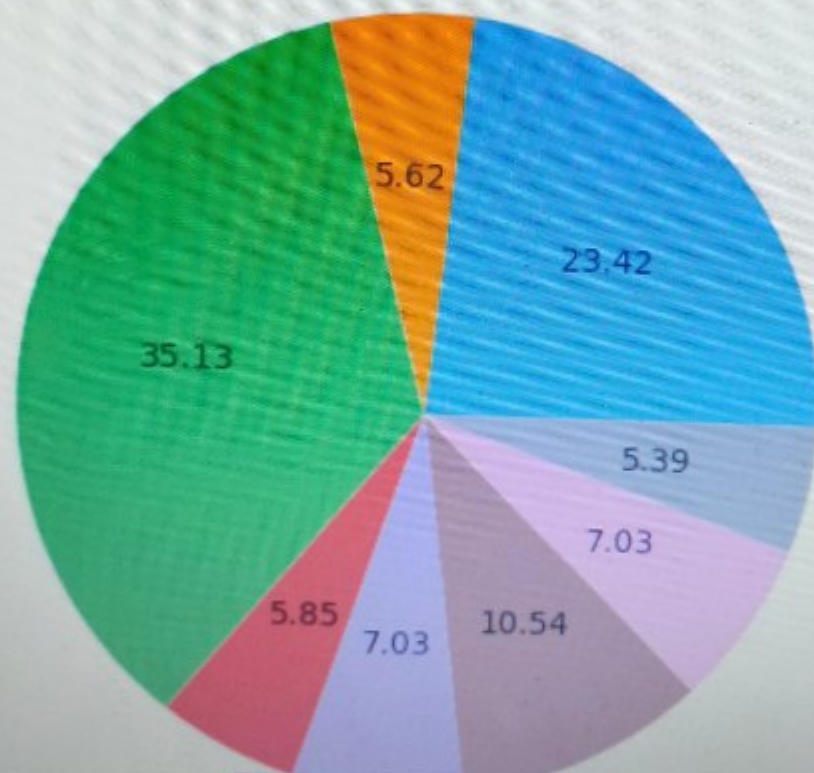


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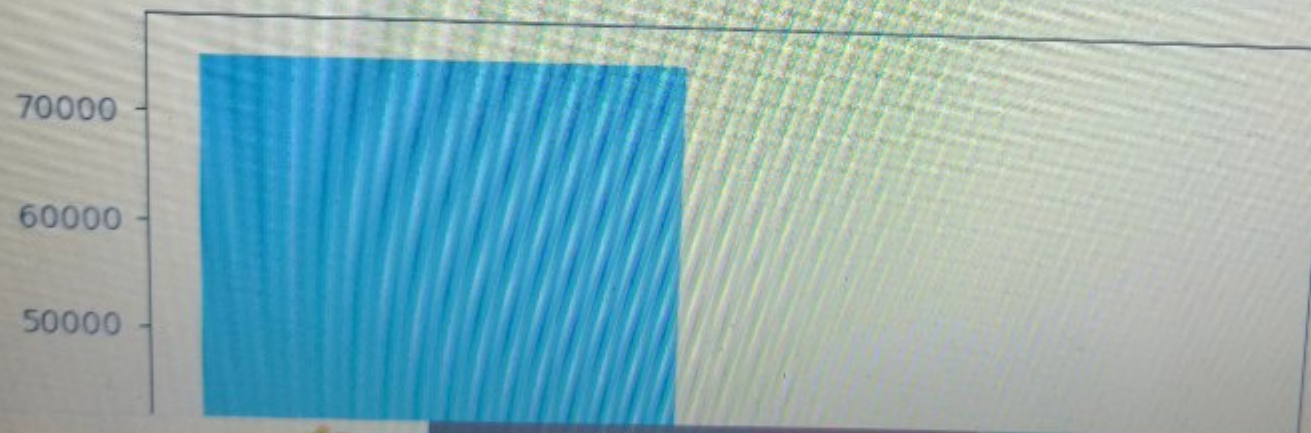
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[63]: # barplot

```
hr_mean = sum(hr_sal)/len(hr_sal)
dev_mean = sum(dev_sal)/len(dev_sal)
```

```
[64]: plt.bar(["HR" , "dev"], [hr_mean , dev_mean])
      plt.show()
```



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Code

HR

dev

```
[65]: # 12 . Multivariate Analysis ; numerical ; num : num
```

```
[66]: df.head()
```

```
[66]:
```

	Salary	dept	Age
0	25000	Hr	20
1	30000	dev	22
2	24000	Finanace	23
3	50000	HR	24
4	23000	fullStack	23

```
[67]: df["Exp"] = [1,2,3,1,2,3,2,1,2]
```



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```
[67]: df["Exp"] = [1,2,3,1,2,3,2,1,2]
```

```
[69]: df.head()
```

```
[69]:
```

	Salary	dept	Age	Exp
0	25000	Hr	20	1
1	30000	dev	22	2
2	24000	Finanace	23	3
3	50000	HR	24	1
4	23000	fullStack	23	2

```
[75]: # 13. Bubbleplot
```

```
plt.scatter(df["Age"], df["Salary"], s = df["Exp"]*20, color = "orange")
```



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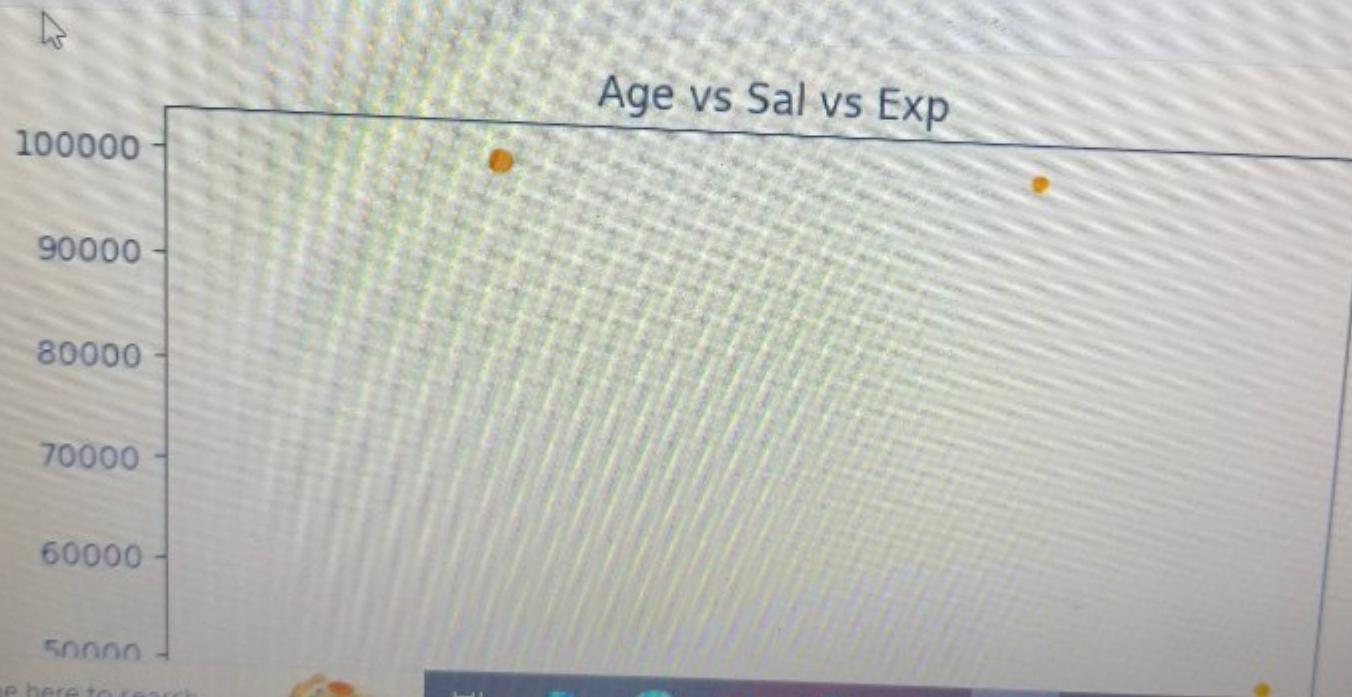
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[71]: # 13. Bubbleplot

```
plt.scatter(df["Age"], df["Salary"], s = df["Exp"]*20, color = "orange")  
plt.title("Age vs Sal vs Exp")  
plt.show()
```



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Code

```
[72]: # 14 . Multi : 2 num : 1 cat

[73]: # scattor plot : self

[74]: # 15. Object oriented API

[78]: fig, axs = plt.subplots(1,3, figsize = (15,5))

      # Line plot

      axs[0].plot(sort_age["Age"], df["Salary"], color = "red")
      axs[0].grid()
      axs[0].set_title("Lineplot")
      axs[0].set_xlabel("Age")
      axs[0].set_ylabel("Salary")
```



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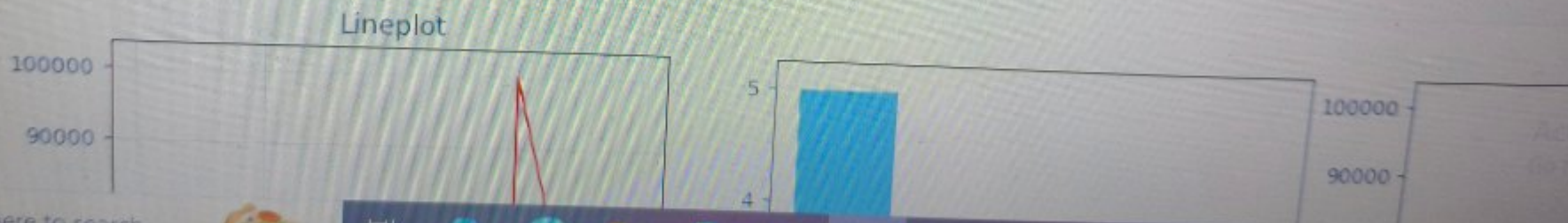
Code

```
axs[0].set_xlabel("Age")
axs[0].set_ylabel("Salary")

#histogram
axs[1].hist(df["Salary"], bins = 5)

#Boxplot
axs[2].boxplot(df["Salary"])

plt.savefig("multipleplot.png")
plt.show()
```



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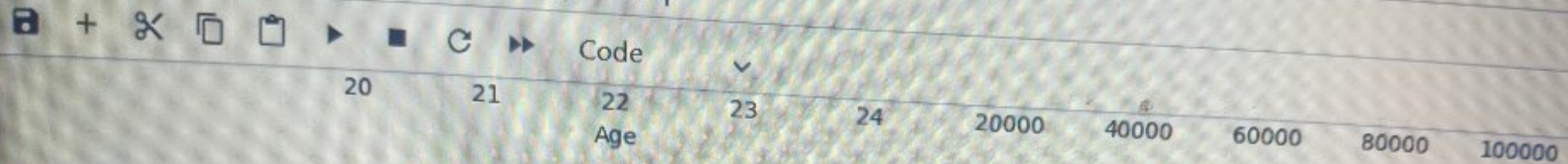
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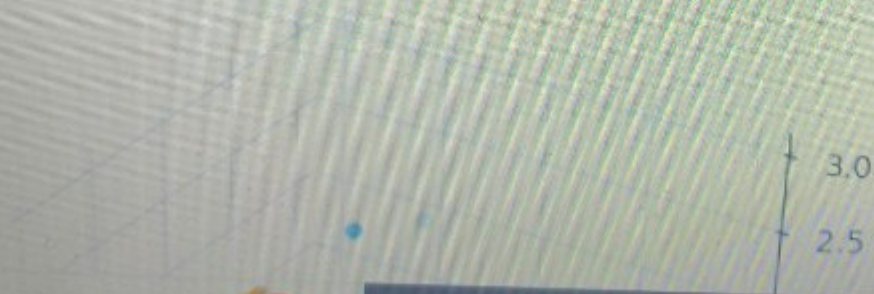
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3D Plot ¶

```
[80]: ax = plt.axes(projection = "3d")  
  
ax.scatter(df["Age"], df["Salary"], df["Exp"])  
plt.show()
```



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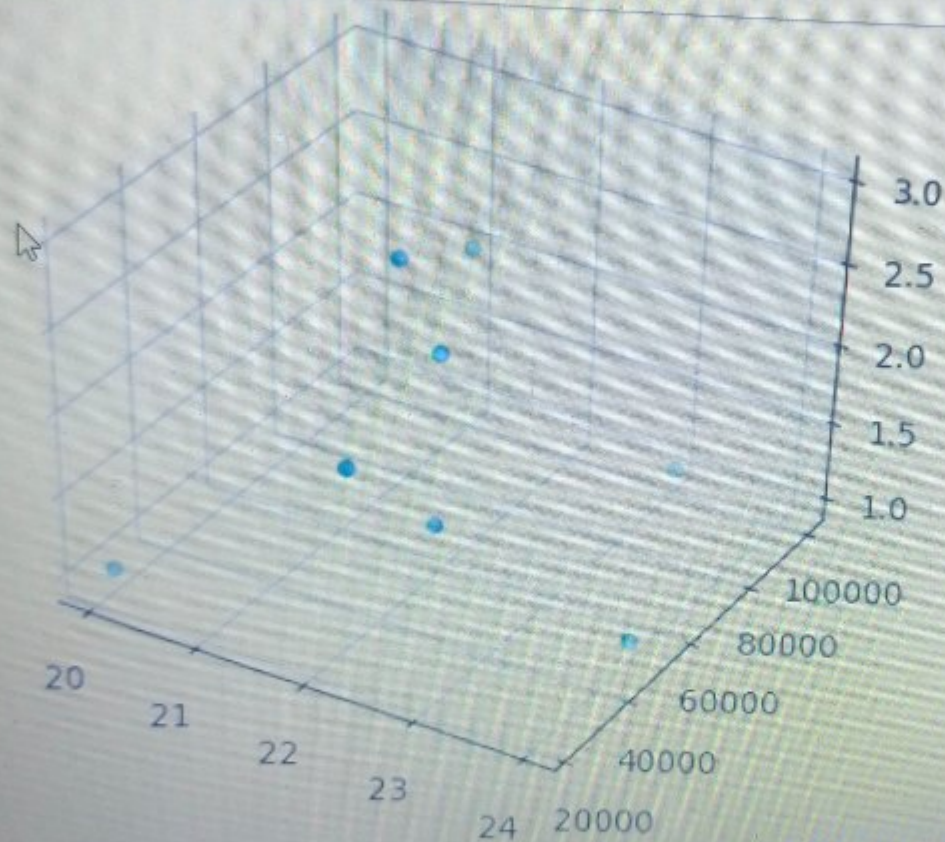
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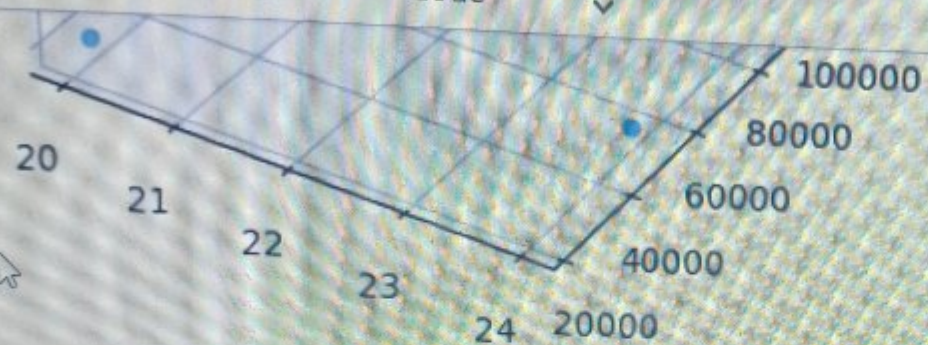


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Code



```
[ ]: #plotly.axis
```



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